

Research Statement

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Foundation models, such as large language models (LLMs), vision language models, and multimodal language models, are highly successful to solve various real-world tasks, such as summarizing documents, answering complex questions, solving math problems, and generating code, images, videos, and stories, as a crucial preliminary step by human towards Artificial General Intelligence (AGI). However, to really achieve AGI, we still need to address many limitations of foundation models, such as **robustness**, **alignment**, **generalization**, and **safe multi-agent decision making**. These issues are important in applications. For example, autonomous driving requires coordination of multiple autonomous vehicles to ensure safe driving performance that is robust to any weather conditions and road types, which involves **robustness** to adversarial environment and **safe multi-agent decision making**. LLMs should provide helpful, honest, and harmless responses via model **alignment** with human values, and **generalize** this ability to a variety of contexts. My research focuses on addressing these important issues using core AI techniques, including large-scale optimization, reinforcement learning, and foundation model alignment.

1 Robust Foundation Models

Foundation model training usually requires large amount of data. However, collecting data from the real world is usually expensive and risky. Therefore, foundation models are often trained in a simulation environment with cheap training data and then deployed to a real-world environment with expensive test data. In this case, it is critical to train a foundation model to make decision that is robust to such an environmental change and thus to preserve good performance when deployed to the real-world environment. My works focus on two major robust learning frameworks to solve this problem, namely, **distributionally robust optimization (DRO)** and **robust reinforcement learning**, for static and dynamic environments, respectively.

1.1 Distributionally Robust Optimization (DRO)

Classical foundation model training typically use empirical risk minimization (ERM), which minimizes the expected loss under the empirical distribution P of the training data and assumes that the test data is also generated from the same distribution P . However, in practice, the test data often follows a different distribution Q . For example, potential adversarial attacks and adding data from the minority group can cause such a distribution shift. Distributionally robust optimization (DRO) problem is an important and emerging generalization of ERM to make the training performance robust to such distribution shift. Many existing theoretical results focus on DRO with convex loss functions, while theoretical results on the more general and commonly used nonconvex loss are scarce and the complexities are very high. Our ICML 2023 work (Chen et al., 2023) for the first time improves the state-of-the-art complexity result of nonconvex DRO algorithm to the corresponding theoretical lower bound under both deterministic and stochastic settings, by proposing a large α -generalized smooth function class that can cover nonconvex DRO, phase retrieval application and language models, etc.

1.2 Robust Reinforcement Learning

Robust reinforcement learning is an extension of standard reinforcement learning, aiming to train a foundation model to make an optimal decision that is robust to any adversarial perturbation of a dynamical environment. For example, an autonomous vehicle should be able to work under any weather conditions and road types, etc. Robust reinforcement learning has been proved NP-hard in general. Hence, many works focus on the tractable s -rectangular case. In this case, despite many empirical successes of policy-gradient based algorithms, theoretical complexity results are rather high and limited to a fully known environment (deterministic case), small state spaces and standard linear utility (i.e., to maximize cumulative reward). Our goal is to improve theoretical efficiency of robust reinforcement learning and extend the results to more realistic situations.

Specifically, one of our ICML 2024 works (Chen and Huang, 2024) improves the state-of-the-art iteration complexity result of policy-gradient based algorithms for s -rectangular robust reinforcement learning, and extends the result from fully known environment (deterministic case) and small state spaces required in the existing theoretical results, to an unknown environment (stochastic case) and large state spaces.

Furthermore, our Neurips 2024 work (Chen et al., 2024a) proposes a new robust reinforcement learning framework with general utility, which for the first time extends robust reinforcement learning from standard linear utility to general utility that can serve a variety of common purposes, including standard cumulative reward maximization (exploitation), exploration maximization, exploitation-exploration trade-off, safe robust exploitation, mimicking expert demonstrations, etc. Then we design two policy-gradient based algorithms for this new learning framework and obtain their theoretical computation complexity results to converge to a stationary solution and a global optimal solution respectively.

2 Foundation Model Alignment and Generalization

As foundation models become more and more widely used, it is essential to align foundation models with human values, and make these models more generalizable.

2.1 Large Language Model Alignment

Large language models (LLMs) including ChatGPT are quite popular foundation models, since they can help human automatically solve a lot of problems, including question and answer, intelligent web search, writing and editing articles and codes, etc. It is important to align LLMs with human preferences to make LLM responses helpful, honest, and harmless. The mainstream alignment techniques are reinforcement learning from human feedback (RLHF) and its simplification called direct preference optimization (DPO) using human preference data. Despite some empirical successes of RLHF and DPO, there are widely recognized challenges to effective alignment. First, human preference data is likely to be **corrupted** due to inexperienced annotators, personal bias, unclear context, and even malicious falsification. Second, LLMs can easily **overoptimize** over the inaccurately estimated reward model and thus yield low-quality responses. Third, LLMs tend to bias towards overly **verbose** (appearing detailed) response. Existing works target at only one of these problems, while few other works require much computation to estimate multiple reward models and lack theoretical guarantee of generalization ability.

In contrast, our current work (Chen et al., 2025) proposes the first RLHF algorithm and its provably equivalent DPO algorithm that simultaneously solve the three issues above (**corruption**, **overoptimization**, **verbosity**), and demonstrates this ability by obtaining generalization theory results. Moreover, in experiments, our DPO algorithm outperforms the vanilla DPO and typical DPO algorithms that target at only one of these issues, with almost as easy implementation as the vanilla DPO.

2.2 Generalizable Minimax Models

Minimax model is fundamental for many machine learning problems, including adversarial training, image generation (via generative adversarial network), AlphaGo, fairness-aware decision making, etc. Existing algorithms for minimax model rely on strong geometric assumptions on the objective function to ensure convergence, such as Lipschitz smoothness and convex-type geometries, which is not generalizable to many machine learning problems. To make minimax models more generalizable, our works obtain theoretical convergence and complexity results of algorithms for minimax models under much weaker geometric assumptions.

Specifically, while existing works require convex-type geometries to guarantee the parameter convergence of gradient descent ascent (GDA) algorithm, the most popular algorithm for minimax model, our ICLR 2021 work (Chen et al., 2021) for the first time generalizes the existing theoretical convergence results to the much more general KŁ geometry satisfied by many practical machine learning problems. Moreover, we obtain various asymptotic convergence rates of GDA algorithm ranging from sublinear convergence up to finite-step convergence, depending on the KŁ geometric parameter. Furthermore, one of our ICML 2024 works (Xian et al., 2024) extends the convergence analysis of minimax model from Lipschitz smooth functions to generalized smooth functions, which include much more applications such as language models, nonconvex distributionally robust optimization (DRO), phase retrieval, etc.

3 Multi-agent and Safe Reinforcement Learning

Many tasks require multiple intelligent agents to cooperate or compete with each other, including robotics, autonomous driving, resource allocation in power networks, role play among large language models (LLMs), etc. How can we coordinate these agents in an efficient and safe way to either reach their collaborative goal or an equilibrium that balances

the competitors’ goals? My works in this area address this issue via **multi-agent reinforcement learning (MARL)** and its extension to **safe MARL**.

3.1 Multi-agent Reinforcement Learning (MARL)

Multi-agent reinforcement learning (MARL) is an emerging learning framework where multiple intelligent agents cooperate or compete with each other in a dynamic environment. Compared with single-agent reinforcement learning, MARL is much more challenging to train. First, each agent faces a non-stationary local environment that is not only affected by the agent’s own action and the unknown environment dynamic, but also the other agents’ actions that are often unpredictable. Second, agents need to collect sufficiently many data from the complex environment and other agents to make wise decision, which may require large amount of computation and communication resources. Third, agents’ private information, such as individual actions and rewards, should be protected in the communication. Our works design fully decentralized privacy-preserving algorithms for MARL with theoretically guaranteed efficiency in data collection and communication.

Specifically, for cooperative MARL where agents collaborate to maximize their averaged cumulative reward, our ICML 2022 work (Chen et al., 2022c) proposes the first privacy-preserving actor-critic and natural actor-critic algorithms with provably efficient sample and communication complexities, which can reduce to the state-of-the-art complexity results in the single-agent case.

For competitive MARL where each agent compete with other agents to pursue its own goal, our ICLR 2022 work (Chen et al., 2022b) focuses on the important special case of two-player zero-sum Markov game with applications to market pricing applications, real-time strategy-making, board games, etc. Specifically, for two-player zero-sum Markov game with unknown environment dynamic, we propose a fully decentralized privacy-preserving stochastic algorithm with almost zero communication cost and much improve the state-of-the-art sample complexity to converge to a Nash equilibrium.

3.2 Safe Multi-agent Reinforcement Learning (Safe MARL)

In risky applications such as autonomous driving and healthcare, AI agents should pursue the optimal performance under safety constraints. Safety is especially important and challenging in MARL since the interaction among the multiple agents may incur additional risks, such as crashes among autonomous vehicles, drones, or robots. The safety requirements along with complex interactions in safe MARL further bring additional challenges to algorithm efficiency. Our aim is to develop algorithms with theoretical guarantees of safety and efficiency for the emerging safe MARL problems, including *constrained cooperative MARL* with cooperative agents and *constrained Markov game* with competitive agents.

Constrained cooperative MARL with cooperative agents has been applied to coordinate drones, autonomous vehicles, directional sensors, etc. Our ICLR 2024 work (Chen et al., 2024b) proves that primal-dual algorithm, the mainstream approach to constrained cooperative MARL, can fail to converge to the global optimal solution since we prove that the duality gap can be nonzero. Instead, we obtain the global convergence rate of the primal-dual algorithm up to the nonzero duality gap. Then we propose a decentralized primal approach and prove its global convergence up to a different constant gap. Moreover, we show that neither of the two algorithms outperforms the other one through numerical examples.

In *constrained Markov game*, agents compete with each other under safety constraints in a dynamic environment, in order to reach a certain safe equilibrium where no one can benefit from changing its own policy alone. Existing works aim to obtain a Nash equilibrium, which has been proved to be a PPAD-complete problem that cannot be solved in polynomial time. Our Neurips 2022 work (Chen et al., 2022a) for the first time extends the tractable correlated equilibrium (CE) notion to constrained Markov games and obtains the first primal-dual algorithm with provable polynomial sample complexity to converge to CE.

4 Future Directions

In the near future, I would like to extend my current research topics to the following directions towards AGI.

- **Efficient Reasoning:** Reasoning ability is crucial towards AGI. Current methods for training foundation models in reasoning ability require to explore a large number of reasoning chains, which is highly costly. Since reasoning is formulated as reinforcement learning and optimization, I can use my expertise in efficiency techniques in these areas to accelerate reasoning, for example, entropy regularization, optimism principle, variance reduction, multi-agent decentralization, etc.
- **Robust Foundation Models to Adversarial Attack:** Foundation models have been demonstrated vulnerable to adversarial attacks, such as jail-break prompts and data poisoning, which are large obstacles to AGI. To train

foundation models to be robust to various adversarial attacks is an important and challenging open problem, which involves my expertise in adversarial training and LLM alignment.

- **Accelerate Large-scale Multi-agent, Safe and Robust Machine Learning.** Many of my works focus on large-scale machine learning with robustness to adversarial environment and multi-agent decision making under safety constraints. Despite existing works on improving efficiency of machine learning algorithms in this area, there is still much space for efficiency improvement. For example, for safe multi-agent reinforcement learning, including constrained cooperative MARL and constrained Markov game, the existing theoretical complexity results are larger than their corresponding lower bounds. My overarching goal is to develop accelerated algorithms for large-scale multi-agent, safe and robust machine learning with theoretical foundation.

In the long term, I would like to explore efficient approaches in the following new and important areas relevant to my research background.

- **Multimodal Learning with Theoretical Foundation.** We are surrounded by a variety of multimodal signals including images, audios, videos, languages, haptic signals, flavors, smells, etc. It is very important for AGI to perceive, understand, and properly respond to these multimodal signals to better serve human needs. A variety of AI models have been recently proposed to learn about these modalities, including multimodal language models, diffusion models, etc. However, theoretical foundation is lacking for multimodal learning. In contrast, for large language models (LLMs), diffusion model, standard reinforcement learning, classical supervised learning, etc., theoretical results have helped improve learning accuracy, reliability and efficiency. For example, the pessimistic principle has been both theoretically proved and experimentally demonstrated to effectively improve generalization ability in both offline LLM alignment and offline reinforcement learning.

Therefore, I am excited to dive into the important and emerging field of multimodal learning theory, to improve accuracy, reliability and efficiency of multimodal learning. A simple way is to extend the theory from LLMs to multimodal signals, since all these multimodal signals including language are usually represented by numeric vectors. My experience in generalization theory of LLM alignment and optimization theory will be helpful in this area.

- **Specific Application Areas.** The ultimate test for AI models (such as foundation models and AGI) lies in applications, such as healthcare, robotics, autonomous driving, power systems, etc. I plan to train safer, more efficient, more robust and value-aligned AI models for these specific application areas, using domain knowledge as well as my expertise in optimization, reinforcement learning, and LLMs.

References

- Chen, Z. and Huang, H. (2024). Accelerated policy gradient for s-rectangular robust mdps with large state spaces. In *International Conference on Machine Learning (ICML)*.
- Chen, Z., Li, J., Yu, P., and Huang, H. (2025). Provably mitigating corruption, overoptimization, and verbosity simultaneously in offline and online rlhf/dpo alignment. *ArXiv:2510.05526*.
- Chen, Z., Ma, S., and Zhou, Y. (2022a). Finding correlated equilibrium of constrained markov game: a primal-dual approach. In *Advances in Neural Information Processing Systems (Neurips)*, pages 25560–25572.
- Chen, Z., Ma, S., and Zhou, Y. (2022b). Sample efficient stochastic policy extragradient algorithm for zero-sum markov game. In *International Conference on Learning Representation (ICLR)*.
- Chen, Z., Wen, Y., Hu, Z., and Huang, H. (2024a). Robust reinforcement learning with general utility. In *Advances in Neural Information Processing Systems (Neurips)*.
- Chen, Z., Zhou, Y., Chen, R.-R., and Zou, S. (2022c). Sample and communication-efficient decentralized actor-critic algorithms with finite-time analysis. In *International Conference on Machine Learning (ICML)*, pages 3794–3834.
- Chen, Z., Zhou, Y., and Huang, H. (2024b). On the hardness of constrained cooperative multi-agent reinforcement learning. In *International Conference on Learning Representations (ICLR)*.
- Chen, Z., Zhou, Y., Liang, Y., and Lu, Z. (2023). Generalized-smooth nonconvex optimization is as efficient as smooth nonconvex optimization. In *International Conference on Machine Learning (ICML)*, pages 5396–5427.

- Chen, Z., Zhou, Y., Xu, T., and Liang, Y. (2021). Proximal gradient descent-ascent: Variable convergence under kl geometry. In *International Conference on Learning Representation (ICLR)*.
- Xian, W., Chen, Z., and Huang, H. (2024). Delving into the convergence of generalized smooth minimax optimization. In *International Conference on Machine Learning (ICML)*.